F70LA exam

December 2019

Solutions

Q1 (a) ${}_3p_{40} = l_{43}/\ l_{40}$

$= 9826.2060 / 9856.2863$

$=$ **0.996948**

(b) ${}_{9|5}q_{40} = (l_{49} - l_{54})/\ l_{40}$

$= (9733.8865 – 9595.9715) / 9856.2863$

$=$ **0.013993**

(c) ${}_6q_{[40]} = 1 - l_{46}/\ l_{[40]}$

$= 1- 9786.9534/9854.3036$

$=$ **0.006835**

(d) $q_{[40]} = 0.000788$

${}_{0.3}q_{[40]} = 0.3 \times 0.000788 = 0.000236$

${}_{0.3}p_{[40]} = 1- 0.000236$

$=$ **0.999764**

(e) $q_{43} = 0.001208$

$p_{43} = 1 - 0.001208 = 0.998972$

${}_{0.8}p_{43} = (0.998972)^{0.8} = 0.999033$

Then use ${}_{3}p_{40}$ from (a) to give ${}_{3.8}p_{43} = 0.996948 \times 0.999033$

$= \mathbf{0.995985}$

(f) Very little importance (1 mark). [Might be important at very high ages, when q is very high.]

Q2. Three conditions: start at 1, finish at 0, non-increasing [but use full definitions].

(a) Yes
(b) No, finishes at -1.
(c) No, dS/dt = 80-2t > 0 initially.

[Full details required for full marks]

Q3 (b) $A_{[40]:20} = .32076$

$D_{60}/D_{[40]} = l_{60} / l_{[40]} / 1.06^{20} = 0.293861$

Value of term [payable at end] = 0.32076 – 0.29386 = 0.026899

Value of term [payable immediately] = $0.026899 * (1.06)^{0.5}$

= **0.027694**

Q4 Value of first 5 years = a bar 5

= a5 x (i/δ)

= 4.4518 x 1.019869

= 4.5403

Then a bar 70 = (a due)$_{70}$ -0.5

= 12.934 – 0.5 = 12.434

EPV of payments after 70 = l_{70}/l_{65} /1.04^5 x 12.434

= 9392.621 / 9703.708 / $(1.04)^5$ x 12.434

= 9.8922

Answer = 4.5403 + 9.8922

= **14.432**

[Note: value of unguaranteed annuity is 14.371]

Q5

(a) Before change…

Value of future benefits = £1m x $A_{50:15}$ = £567,190

Value of future premiums = £4,000 x $a_{50:15}$ = 4000 x 11.253 = £46,092

Reserve = 567,190 – 46,092 = £521,098

Then deduct £500 to get £520,598

After change, value of future premiums = £4,000 x a_{50} = 4000x17.444
= £69,776

And A_{50} = 0.32907

Then new SA = (520598 + 69776) / 0.32907
= **£1,794,068**

(b) May not have been too low originally: looks OK on about 7%.
In any case, it is what it is, so any error at the start is irrelevant.

(c) This is correct.
The sum assured is increasing very significantly, and the customer may have reason to believe that he will die soon.
The office may wish to either request medical information, or simply refuse the requested change.

Q6 (a)

$q_{64} = 0.005073$
$p_{64} = 0.994927$
$a_{65} = 13.666$

At 1/1/2018,

EPV one annuity = £10,000 x 13.666 x 0.994927 / 1.04 = £130,737
EPV one death benefit = £50,000 x 0.005073 / 1.04 = £244

Total excluding expenses = 80 (130727 +244) = £10,478,490

Expenses: abar20 = a20 x (i/δ) = 13.5903 x 1.019869 = 13.8603
PV expenses = 20000x 13.8603 = £277,207

Total reserve = 10478490 + 277207 = **£10,755,697**

(b) At year end, have

10755697x1.022-50000 -30000 = £10,912,312

Reserve for annuities = 79 x £10,000 x 13.666 = £10,796,140
Reserve for expenses = 20000 x 13.1339 x 1.019869 = £267,897
Total reserve = £11,064,037

Loss = 11064037 – 10912312 = **£151,715**

(c) Interest loss = (4% - 2.2%) x 10755697

= **£193,603**

Mortality gain = (1- 80 x 0.005073) x (£10,000 x 13.666 -50000)

= **£51,490**

Expenses loss = 30000 -20000x 1.019869 = **£9603**

[note: total loss = £193,603 - £51,490 + £9,603 = £151,715 as in (b)]

(d)

- The basis may be confidential
- It is unprofessional to speak of an actuary in this way
- Should possibly make an allowance for inflation
- And consider expenses declining as annuitants die
- But expenses are only about 3% of the reserve, so greater accuracy is probably unnecessary

Notes for external examiner

1) (a) to (e) are standard. (f) is unseen.
2) Standard.
3) Standard.
4) This has been covered in a tutorial.
5) (a) is standard. (b) is unseen. (c) is an application of principles that have been discussed.
6) (a) is standard for annuities. The expense basis is unseen but is an application of a previous year's course. (b) is difficult but has been covered. (c) is more challenging still but has been covered. (d) is unseen.